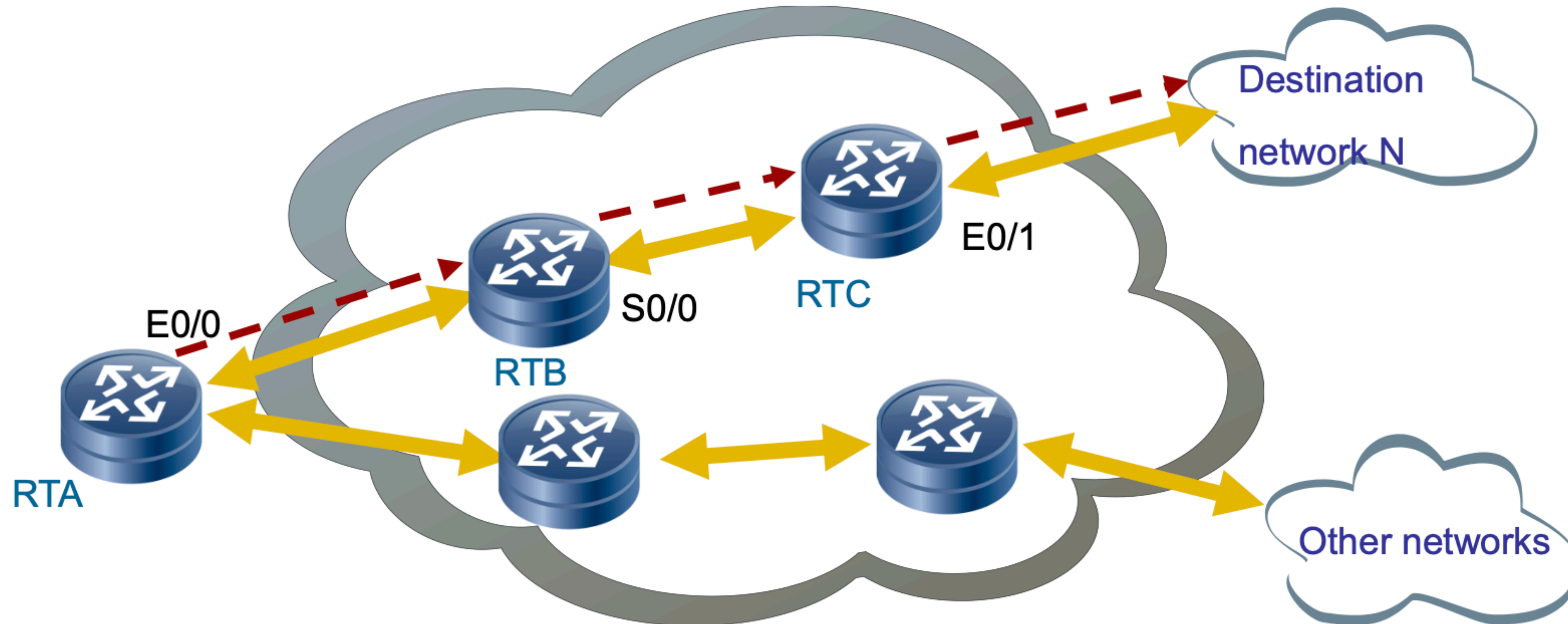


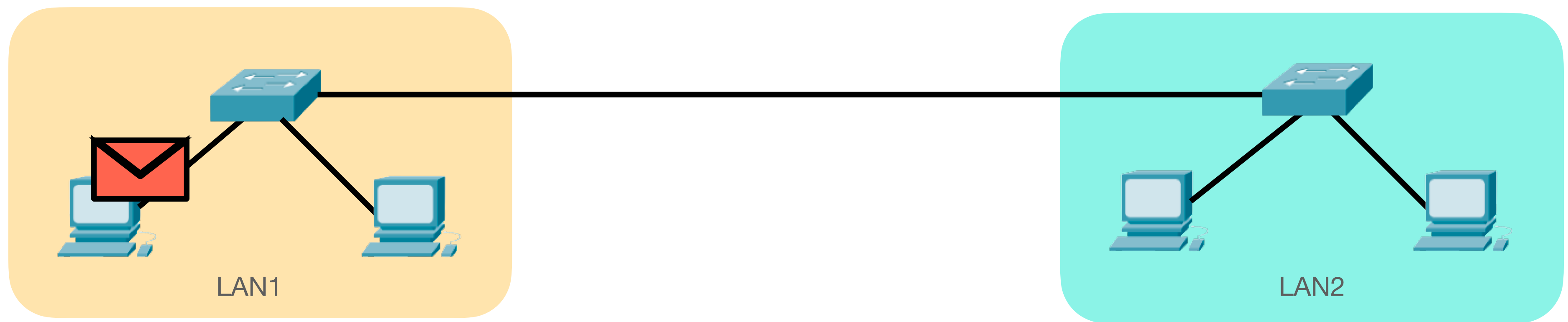
Chapter 5

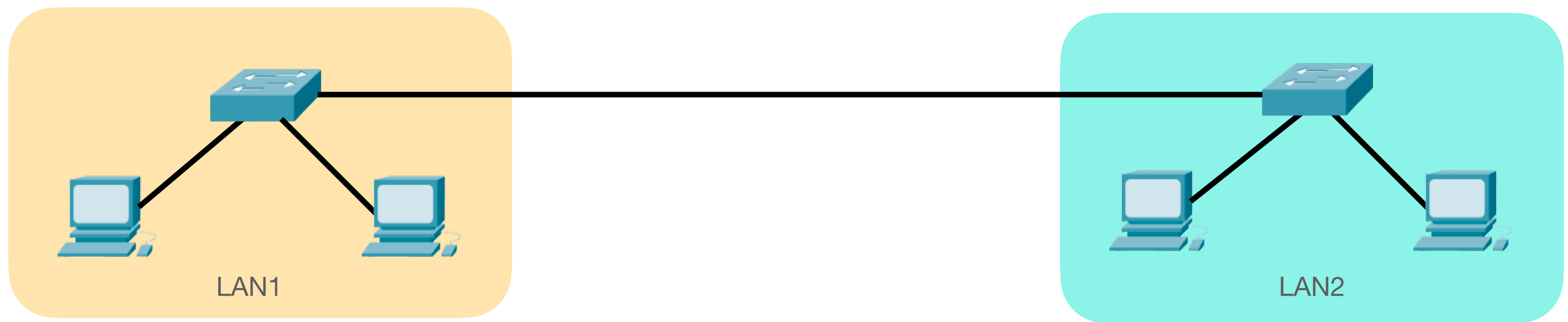
Network Routing

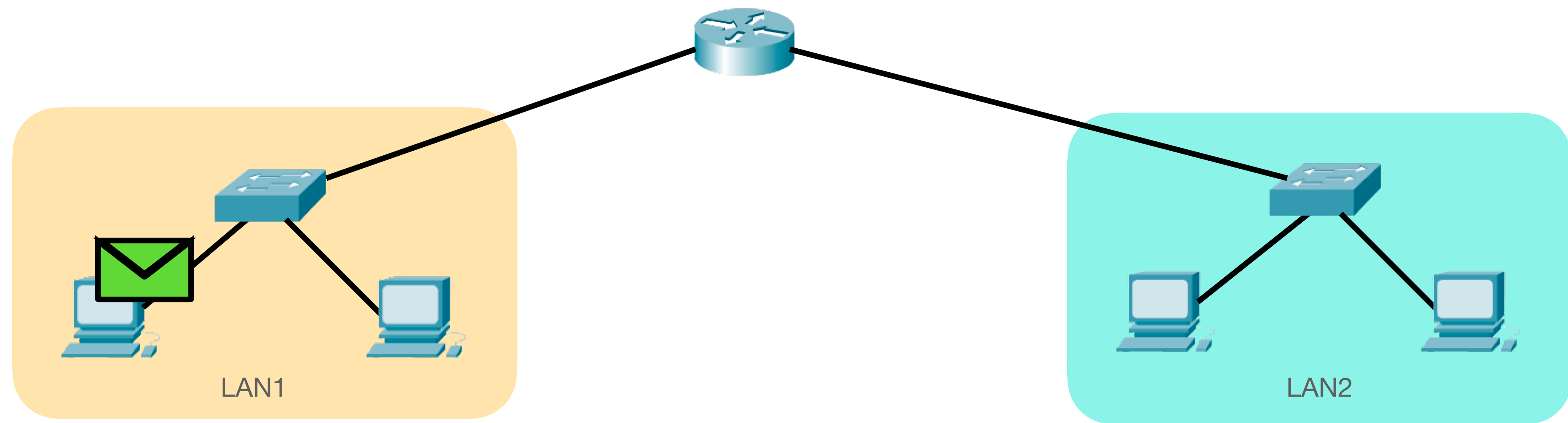
What's The Route?

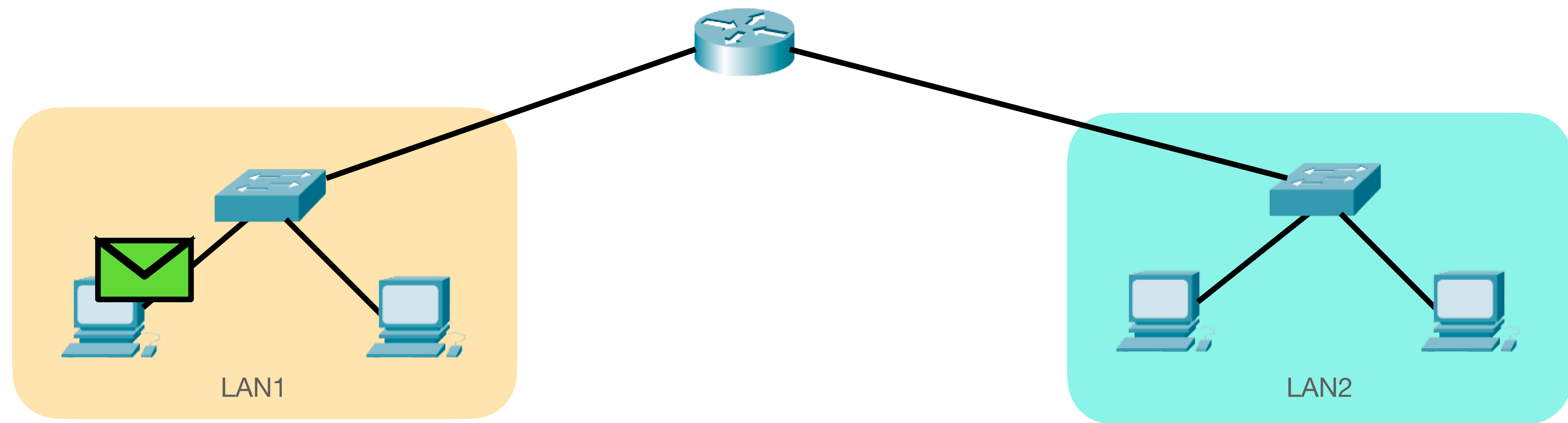
- A route is the predetermined forwarding path used to forward IP packets.





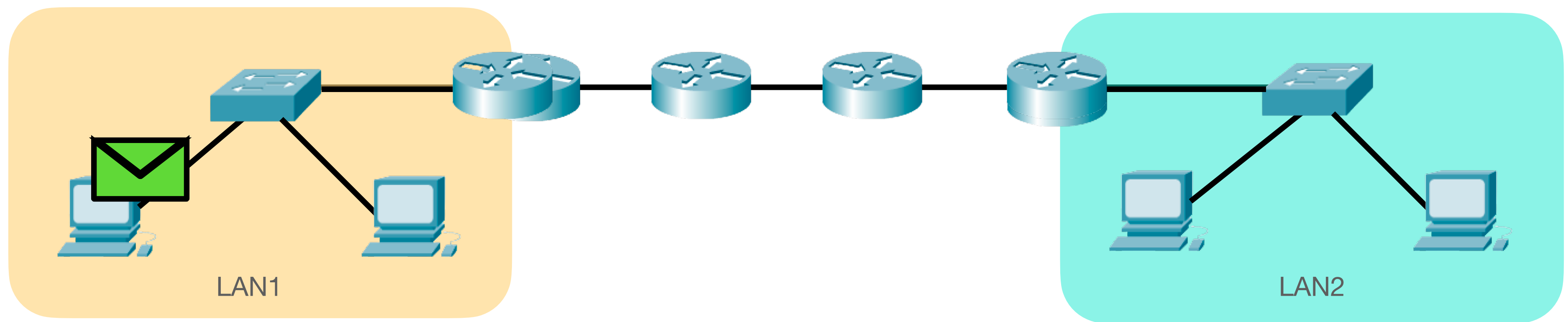


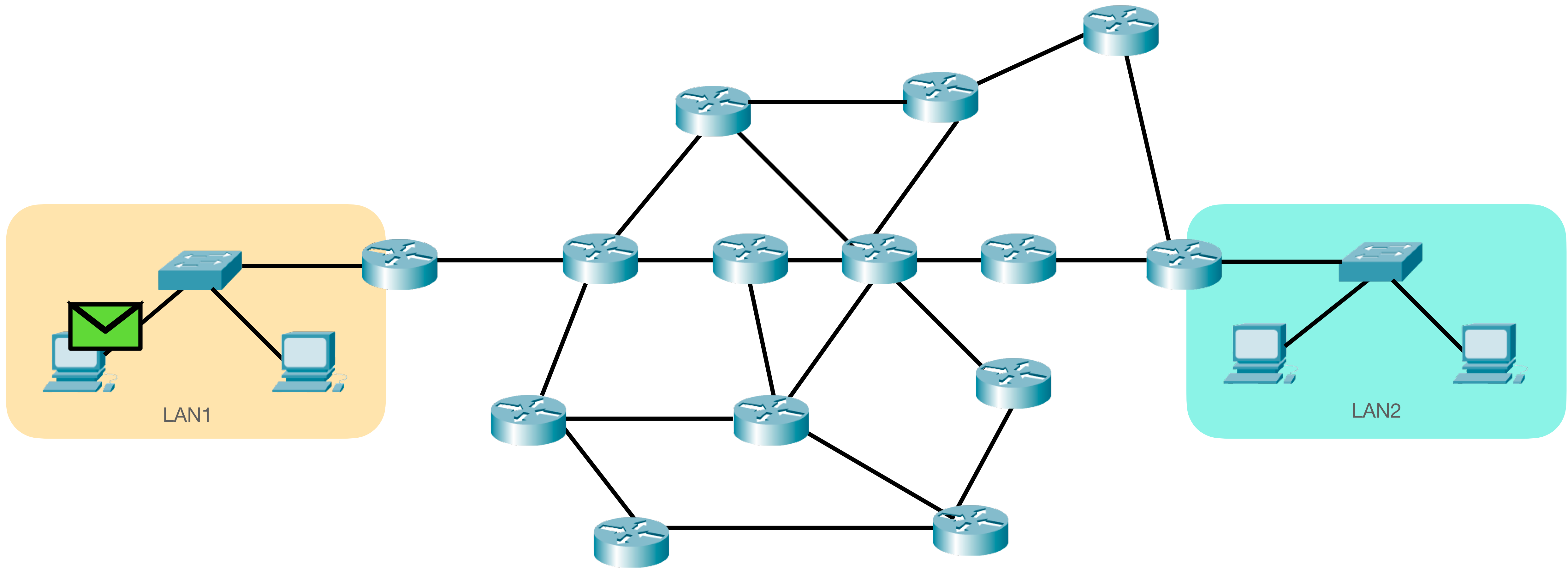










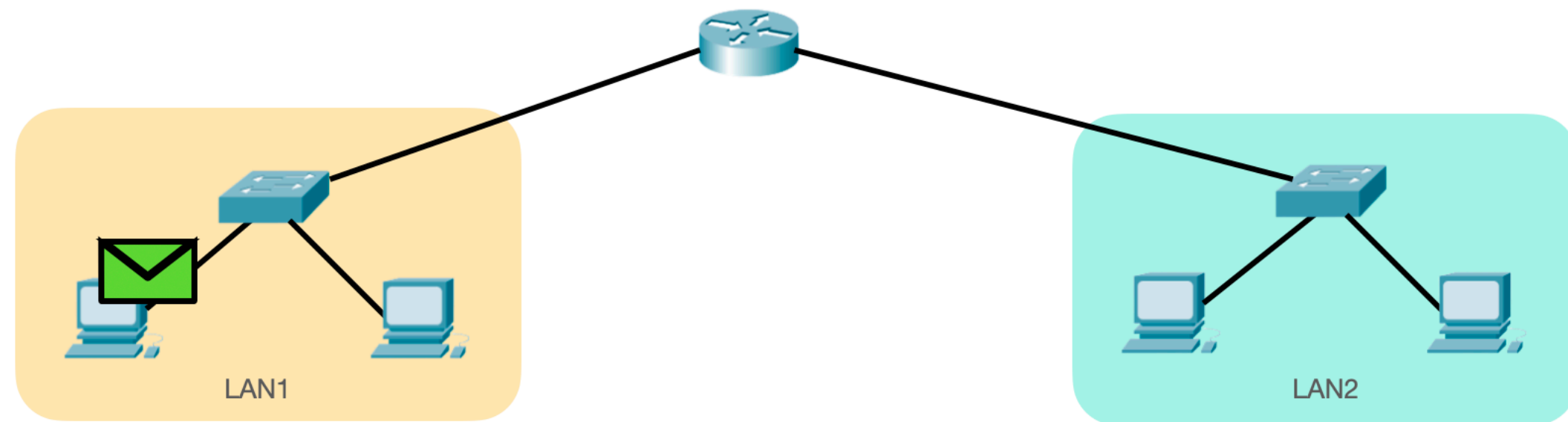


LAN1

LAN2

When is routing needed?

We need to configure routings if there are two or more routers



No routing needed



Routing is needed

Router

Router only has ONE job



To forward packet to a destination network

Using the best path

Based on the routing table

And to reply an ICMP error when data is not sent

Routing Process in Router

- A router must perform the following steps while making routing decision:
 - The router receive data (with IP address, both source IP and destination IP) from an interface.
 - The router then checks destination IP of the data to see if the destination network address exists in its routing table.
 - If Yes: From the routing table, the router determines which interface to use to forward the packet.
 - If no, the router will discard the packet and send an ICMP destination network unreachable message to the source of the packet.
 - The packet continues this process until it reaches its destination.

Types of Routing

Two Types of Routing

Static route

- The route configured manually by network administrator
- When network topological is changed, administrator need to update static route

Dynamic route

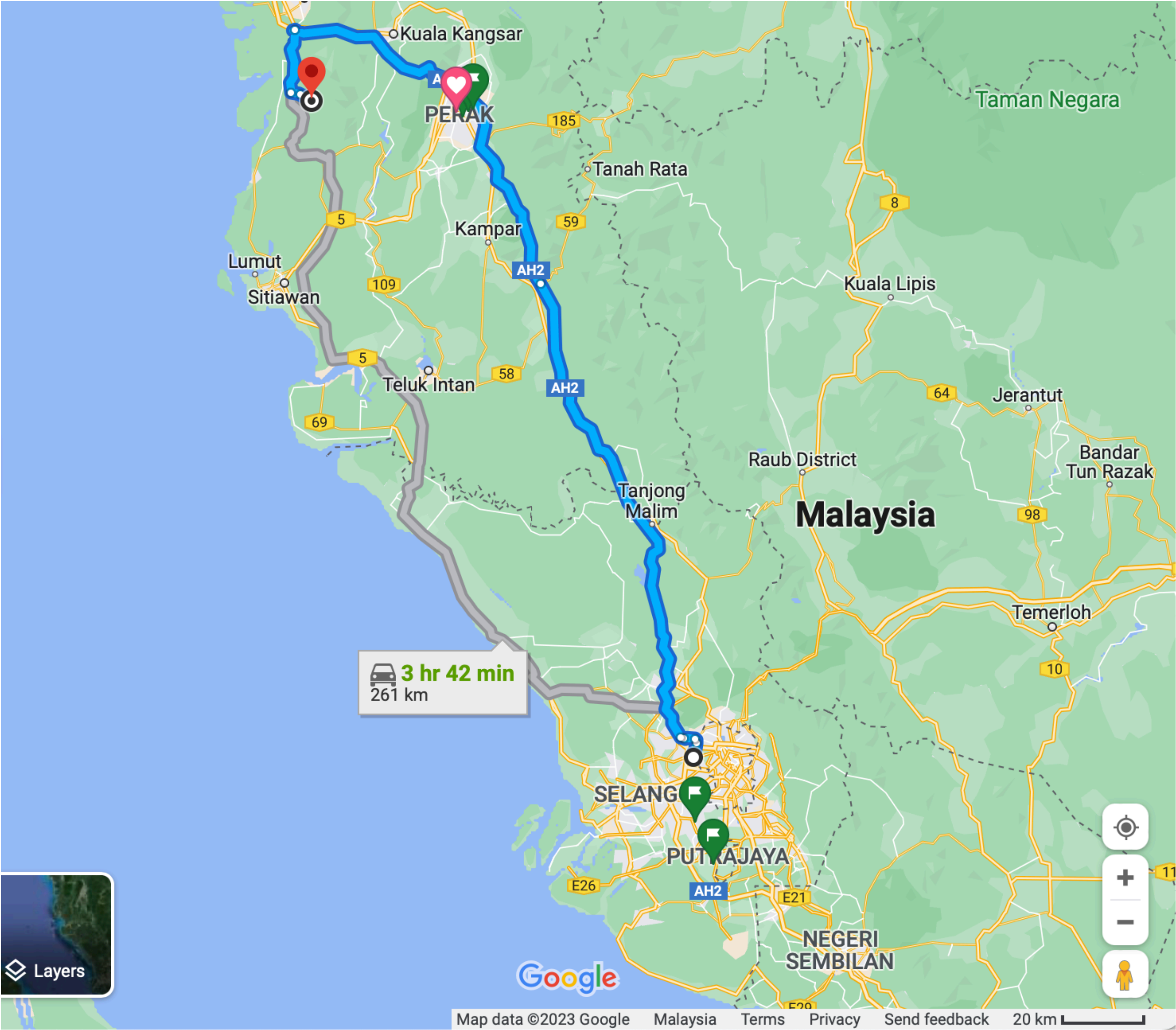
- The route that router gets from other routers via routing protocol
- When network topological is changed, routers will update route information automatically

Distance
Vector

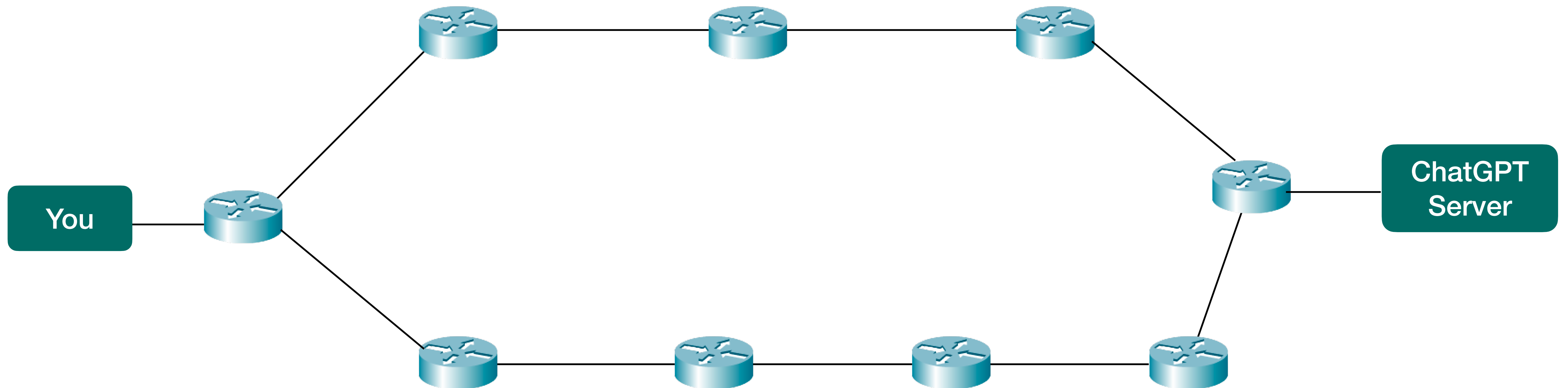
Link State

Best Routes

Best Path
(Route Selection)

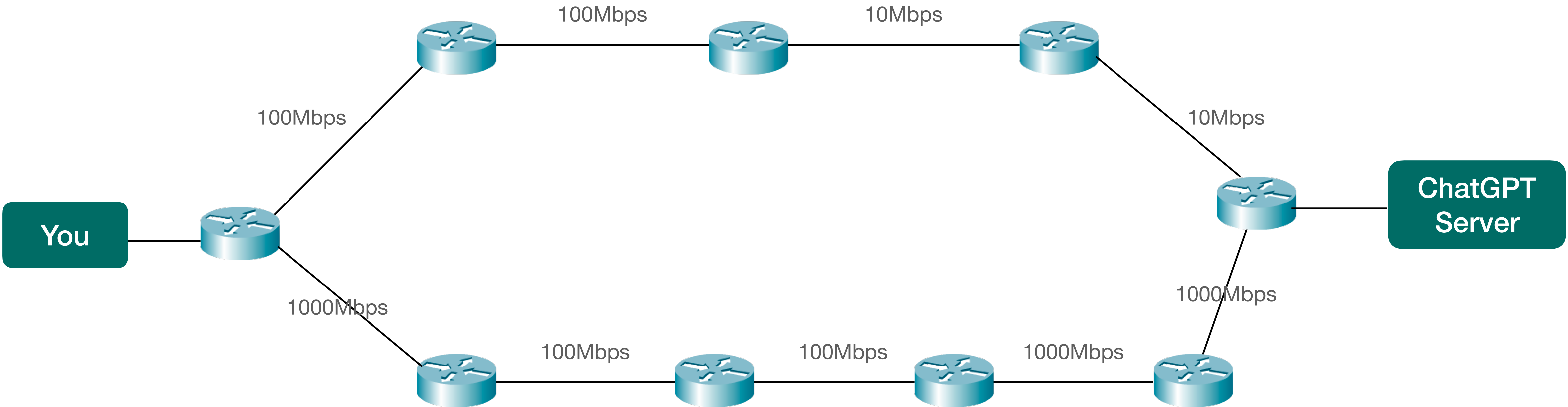


Best Path 14.0x
(Route Selection)



Distance Vector

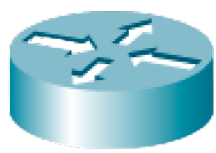
**Best Path 14.0x
(Route Selection)**



Link State

Routing Table

Routing Table



Where the packet is going to?

Who is the next router that will forward your packet?

Which interface on the router should the data exit from?

Destination Network	Next Hop IP	Exit Interface
1.1.1.0	1.1.1.2	fa0/0
2.2.2.0	2.2.2.2	fa0/1
3.3.3.0	3.3.3.1	Fa1/0
4.4.4.0	4.4.4.1	Fa1/1

Routing Table



```
2501A#sh ip route
```

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default U - per-user static route, o - ODR, P - periodic downloaded static route T - traffic engineered route

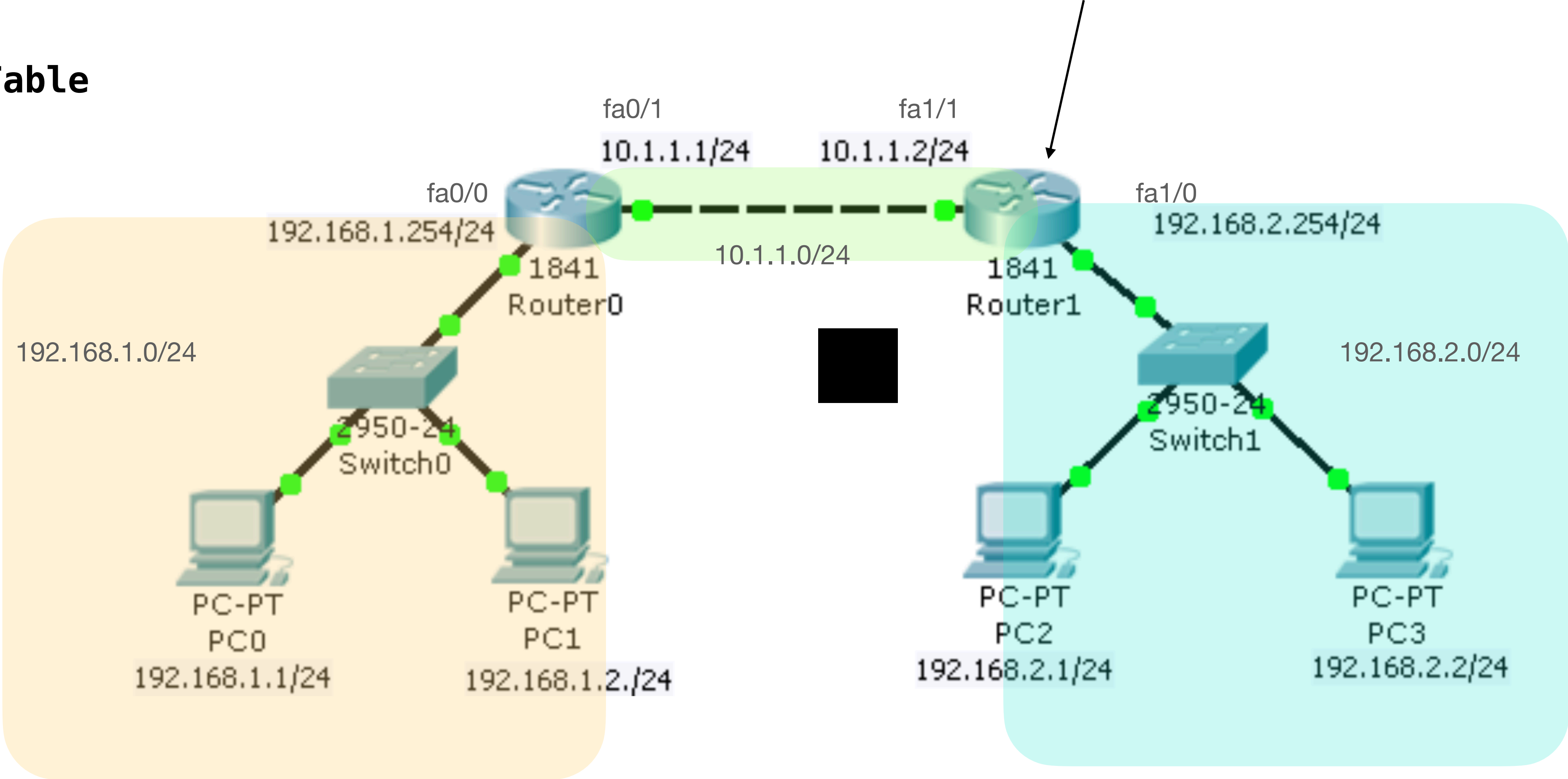
Gateway of last resort is not set

```
172.16.0.0/16 is subnetted, 1 subnets
C      172.16.50.0 is directly connected, FastEthernet0/0
C      192.168.24.0 is directly connected, FastEthernet0/1
10.0.0.0/8 is subnetted, 1 subnets
C      10.10.10.0 is directly connected, Serial0/0
R      175.21.0.0/16 [120/1] via 10.10.10.1, 00:00:18, Serial0/0
```

```
R      175.21.0.0/16 [120/1] via 10.10.10.1, 00:00:18, Serial0
```

- R
 - A code indicating how the route entry was learned on this router. In this case, the R stands for RIP (a form of dynamic routing).
- 175.21.0.0/16
 - The network address and prefix length (number of bits set to 1 in the subnet mask) of the destination network.
- [120
 - The administrative distance of the route.
- /1]
 - The metric of the route specific to the routing protocol used to determine the route.
 - RIP uses hops as its metric. In this example, there is one router between this router and the destination.
 - Different routing protocols have different set of metrics
- via 10.10.10.1
 - The next-hop address (gateway) for the route.
 - This is the IP address that the packet will exit from the LAN in order for the packet to reach its destination.
- 00:00:18 
 - The length of time since the route has been updated in the routing table. In this example, the route was updated 18 seconds ago.
- Serial0
 - The interface the route was learned through.
 - This is also the interface the packet will be switched to in order for the packet to be forwarded toward its destination.

Routing Table



Dest Net	Next Hop IP	Exit Interface
192.168.1.0	-	Fa0/0
10.1.1.0	-	Fa0/1
192.168.2.0	10.1.1.2	Fa0/1

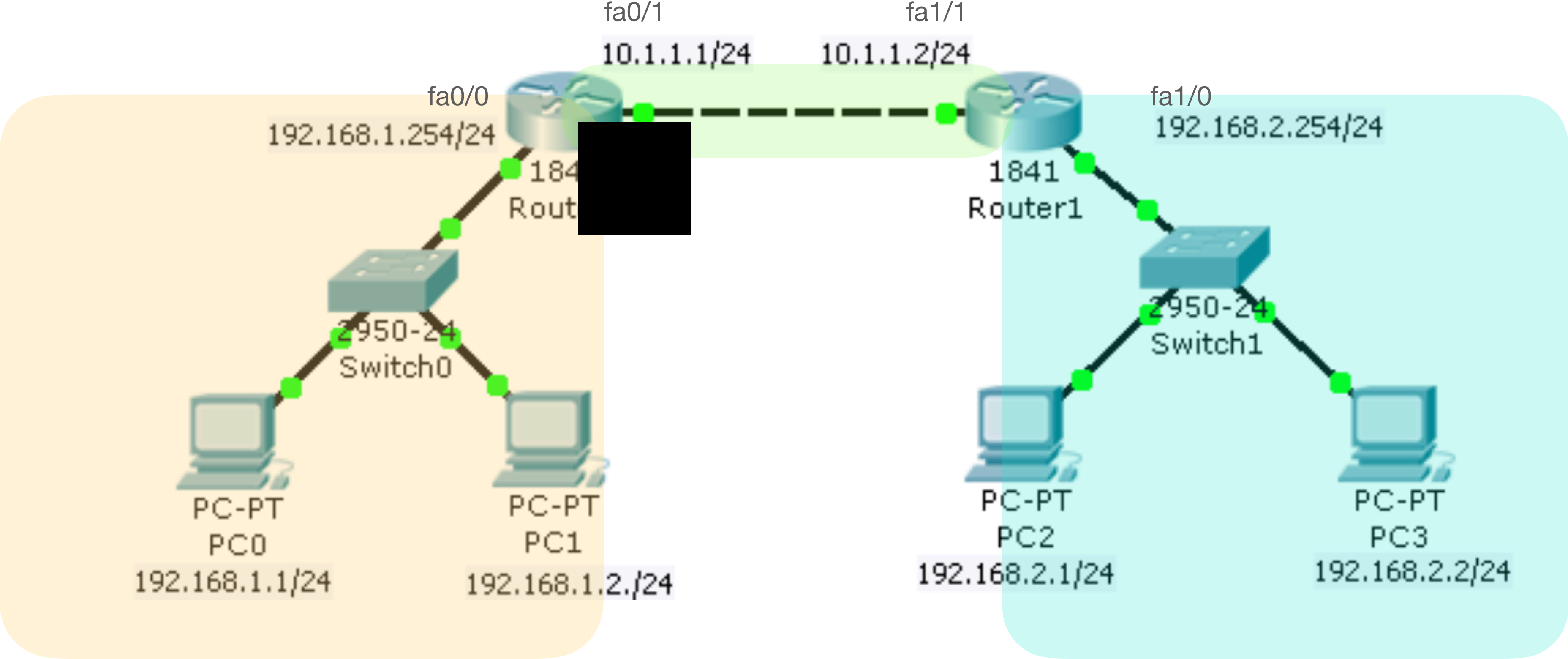
Router0's Table

Dest Net	Next Hop IP	Exit Interface
192.168.2.0	-	Fa1/0
10.1.1.0	-	Fa1/1
192.168.1.0	10.1.1.1	Fa1/1

Router1's Table

Static Routing

How to set Static Routing?



Dest Net	Next Hop IP	Exit Interface
192.168.1.0	-	fa0/0
10.1.1.0	-	fa0/1
192.168.2.0	10.1.1.2	fa0/1

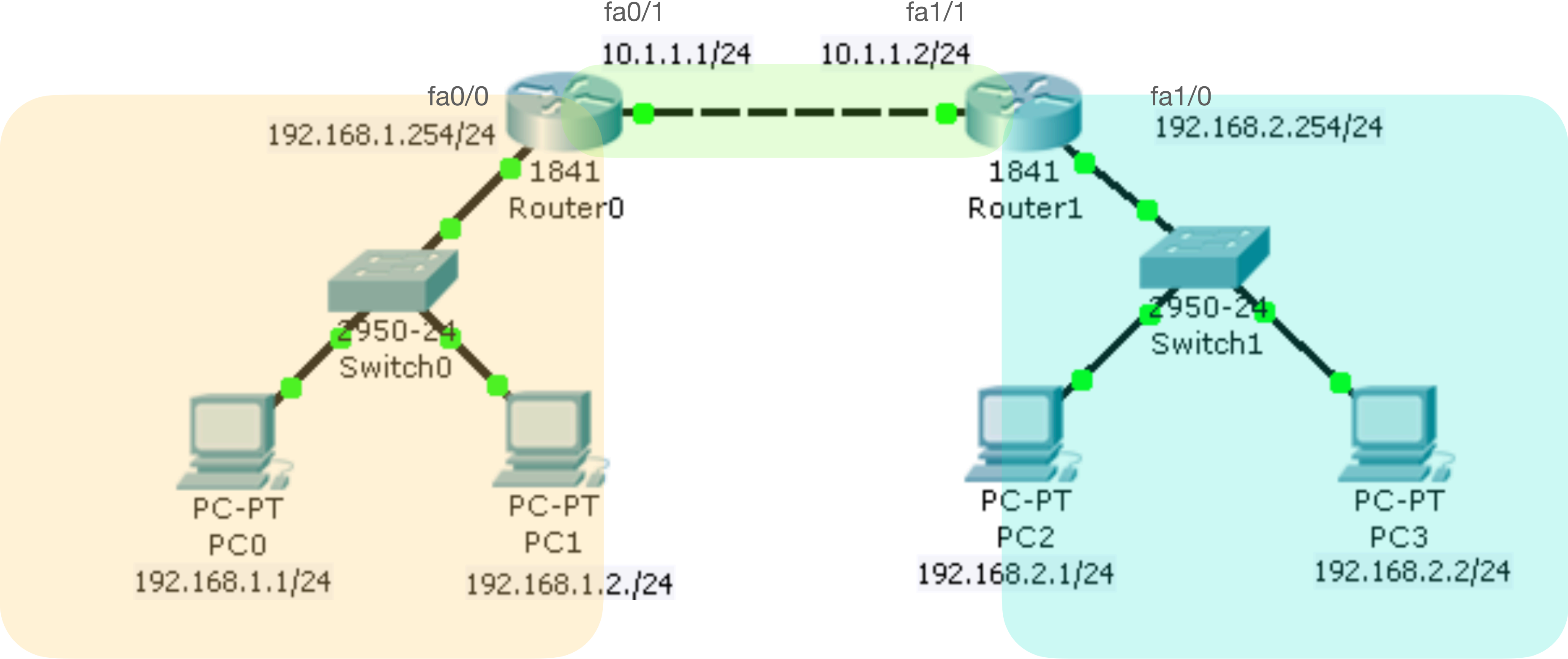
Router0's Table

Router0

```
ip route [dest net] [next hop]
```

```
ip route 192.168.2.0 255.255.255.0 10.1.1.2
```


How to set Static Routing?



Router1

```
ip route [dest net] [next hop]
```

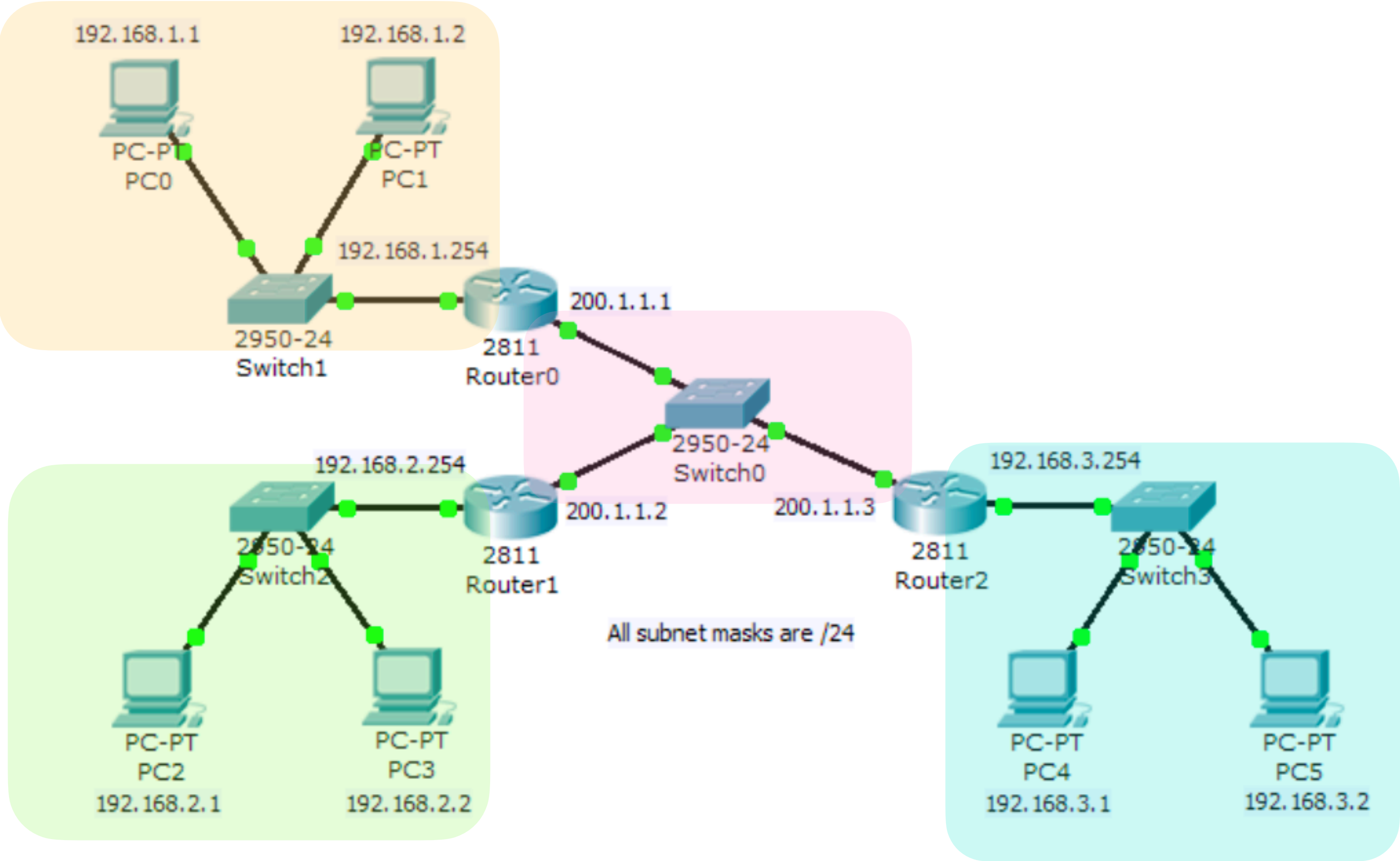
```
ip route 192.168.1.0 255.255.255.0 10.1.1.2
```

Dest Net	Next Hop IP	Exit Interface
192.168.1.0	10.1.1.1	fa1/1
10.1.1.0	-	fa1/1
192.168.2.0	-	fa1/0

Router1's Table

Example 2

Routing Table

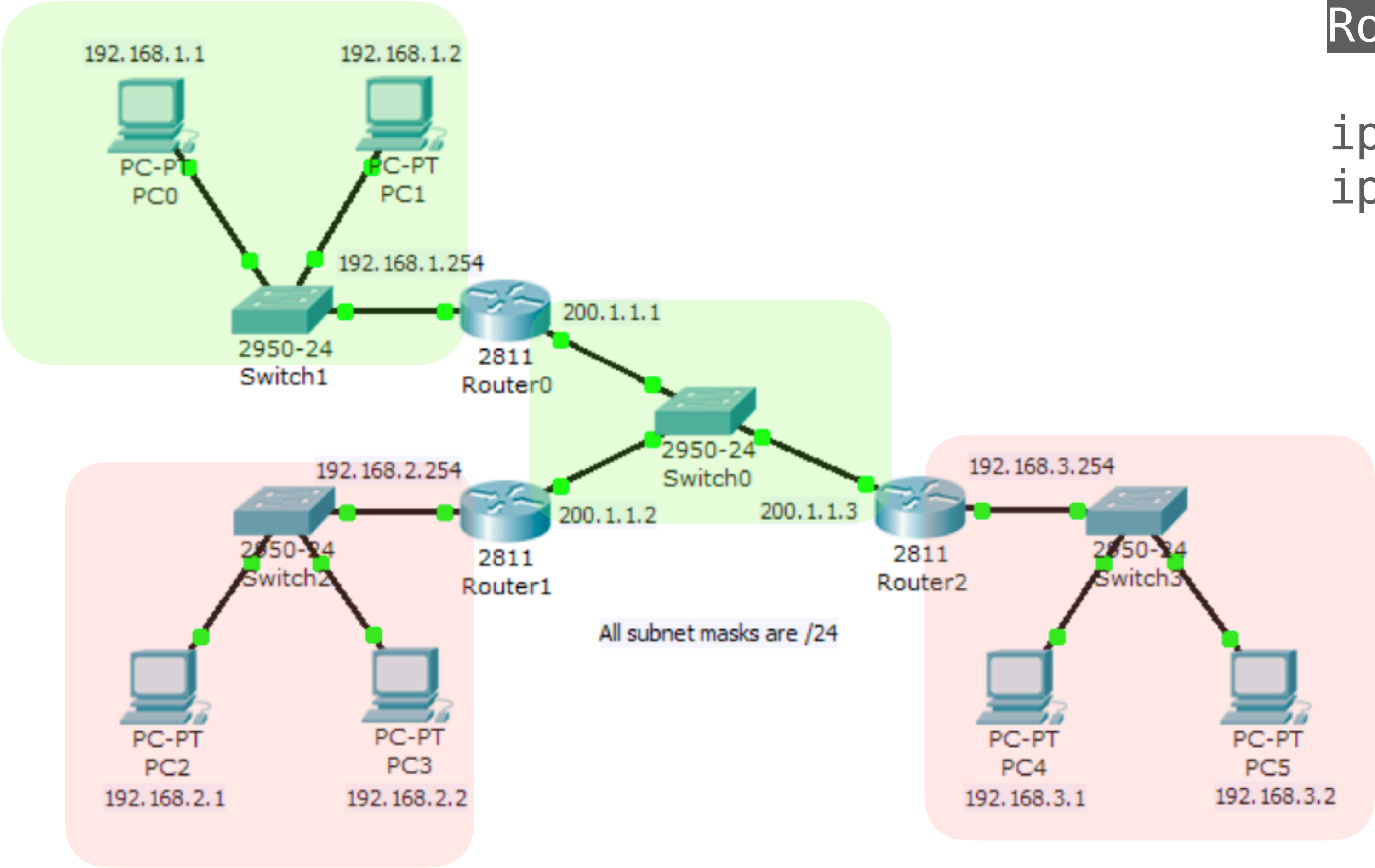


Router0

Router1

Router2

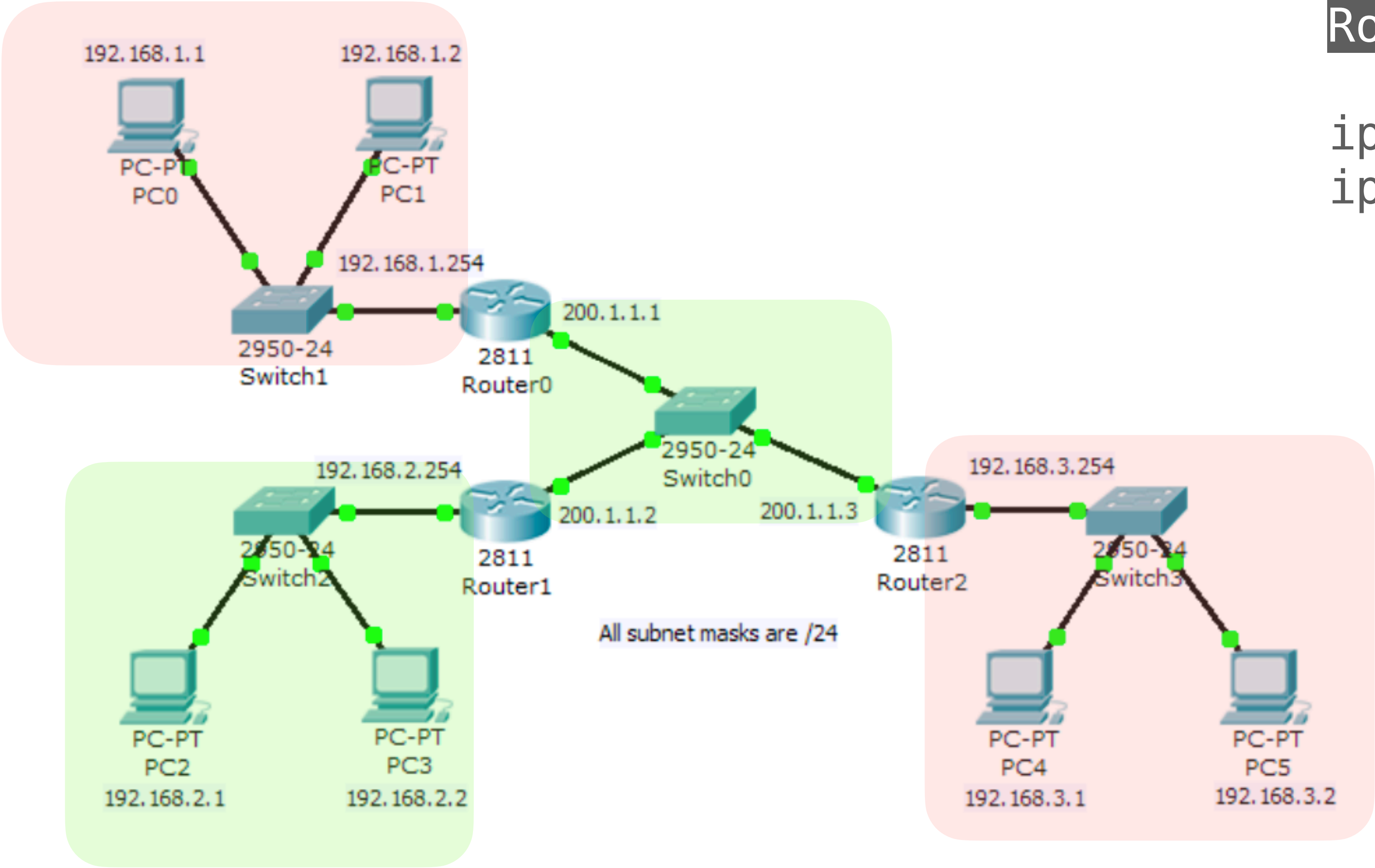
Static Route



Router0

```
ip route 192.168.2.0 200.1.1.2
ip route 192.168.3.0 200.1.1.3
```

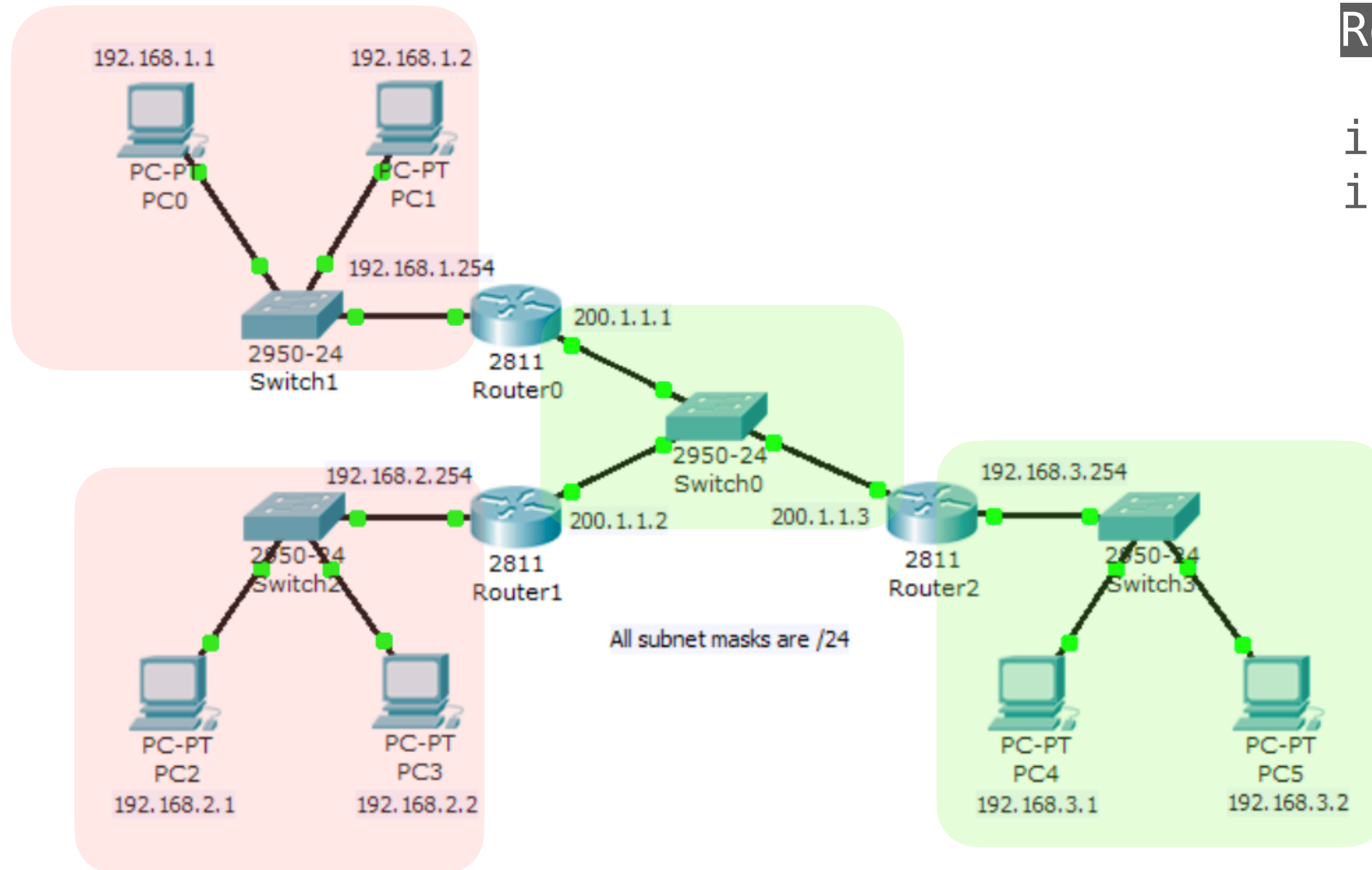
Static Route



Router1

```
ip route 192.168.1.0 200.1.1.1
ip route 192.168.3.0 200.1.1.3
```


Static Route



Router2

```
ip route 192.168.1.0 200.1.1.1  
ip route 192.168.2.0 200.1.1.2
```

Static Route

Router0

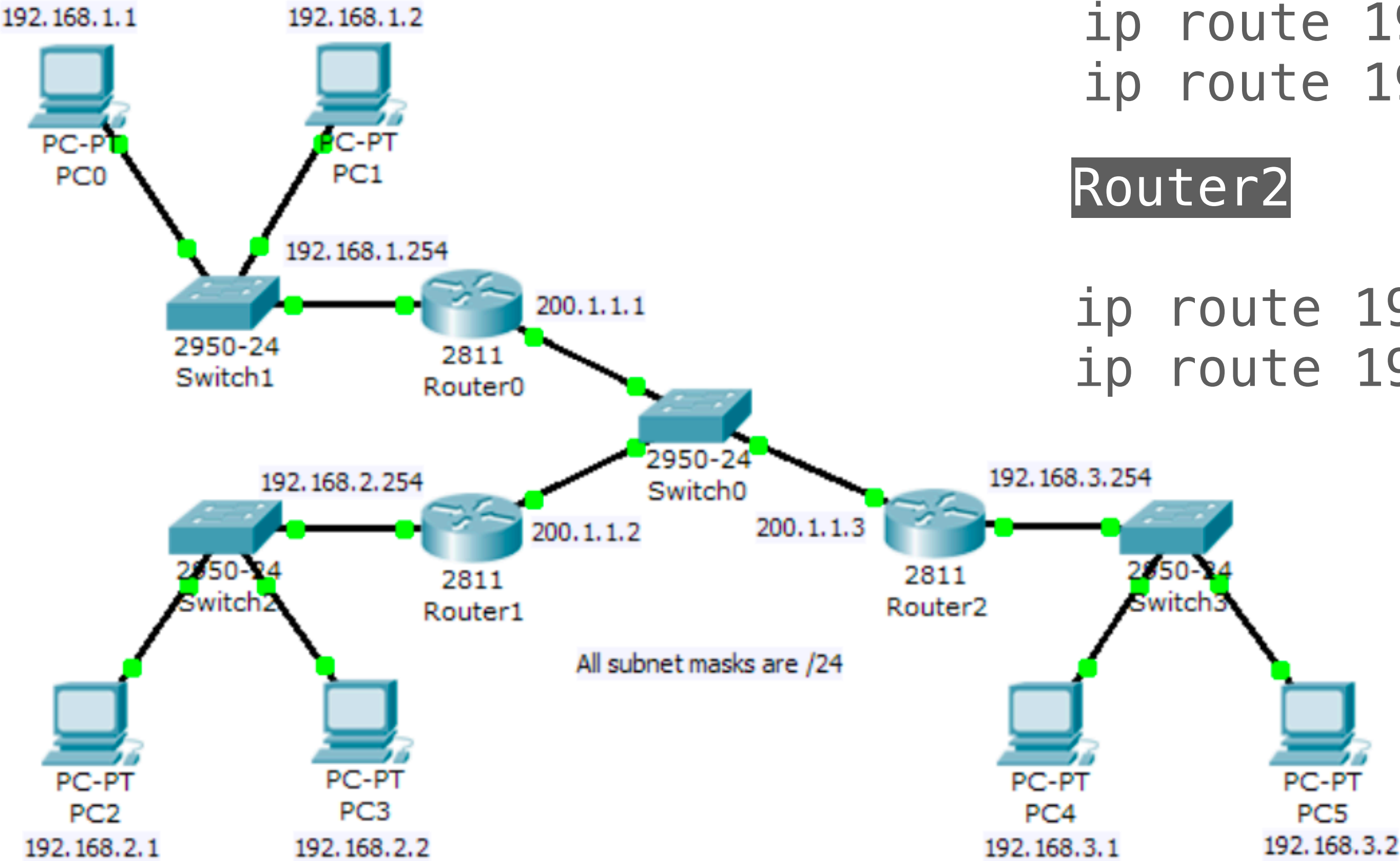
```
ip route 192.168.2.0 255.255.255.0 200.1.1.2
ip route 192.168.3.0 255.255.255.0 200.1.1.3
```

Router1

```
ip route 192.168.1.0 255.255.255.0 200.1.1.1
ip route 192.168.3.0 255.255.255.0 200.1.1.3
```

Router2

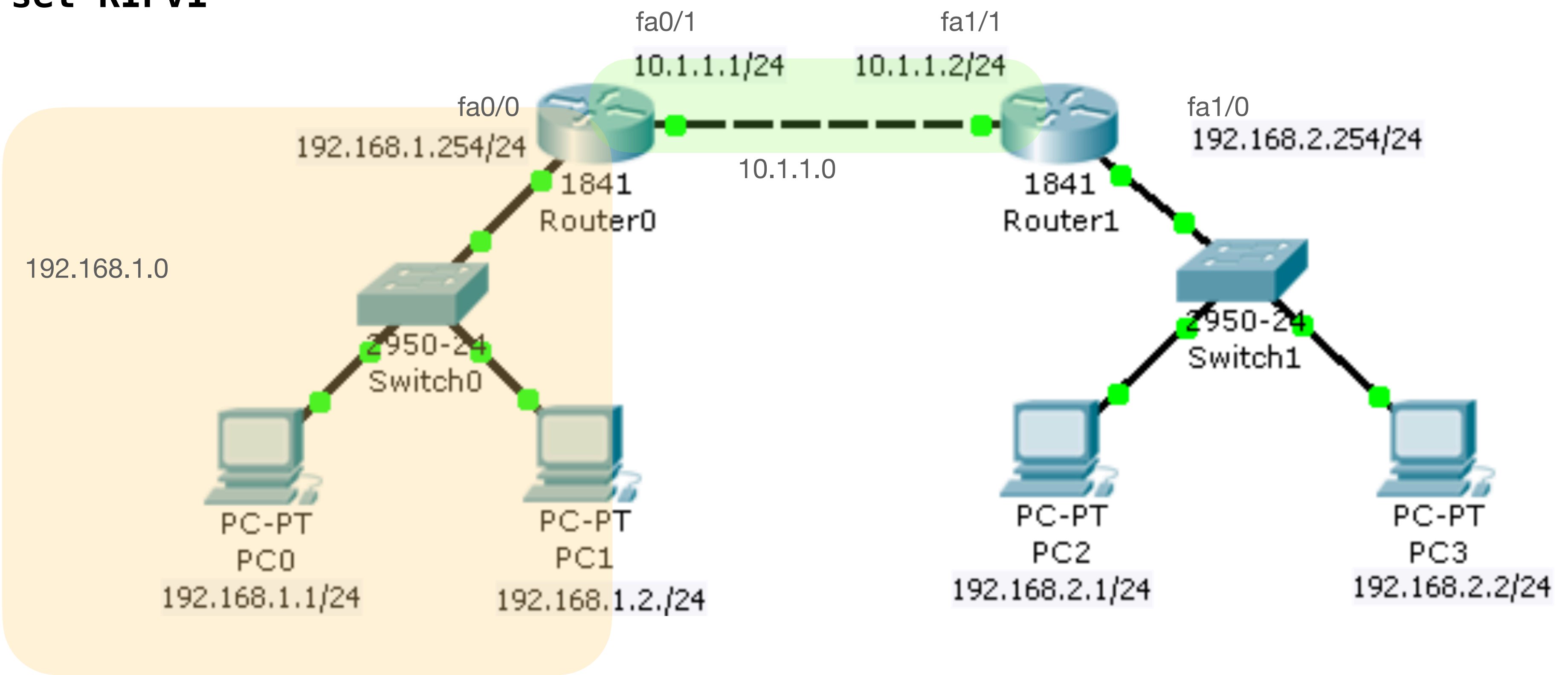
```
ip route 192.168.1.0 255.255.255.0 200.1.1.1
ip route 192.168.2.0 255.255.255.0 200.1.1.2
```



Dynamic Routing

(RIPv1)

How to set RIPv1



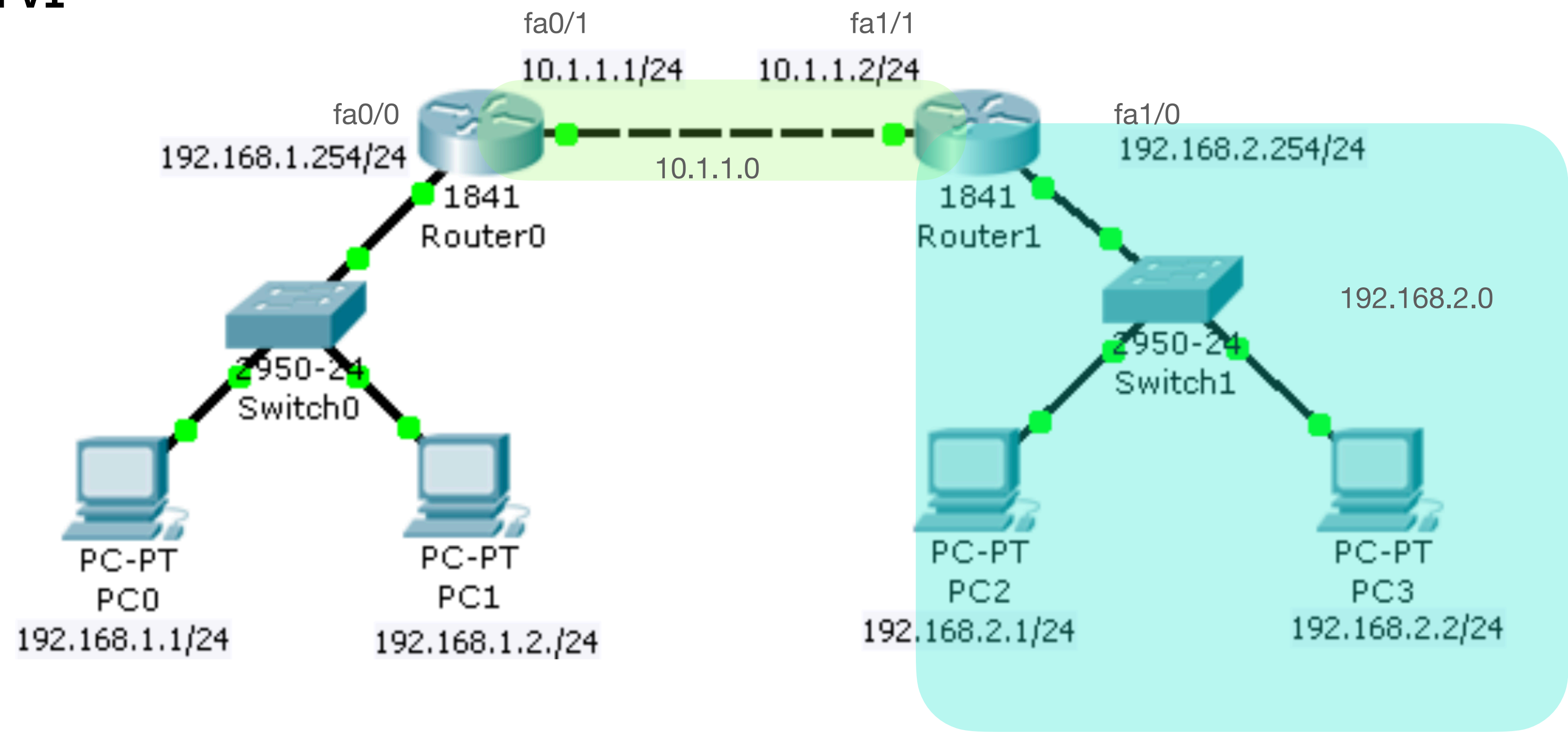
Types	Dest Net	Next Hop IP	Exit Interface
C	192.168.1.0	-	fa0/0
C	10.1.1.0	-	fa0/1
R	192.168.2.0	10.1.1.2	fa0/1

Router0's Table

```
router rip
network [network ID]
```

```
router rip
network 192.168.1.0
network 10.1.1.0
```

How to set RIPv1



Types	Dest Net	Next Hop IP	Exit Interface
R	192.168.2.0	10.1.1.2	
C	10.1.1.0	-	fa1/1
C	192.168.1.0	-	fa1/0

Router0's Table

Types	Dest Net	Next Hop IP	Exit Interface
C	192.168.2.0	-	fa0/0
C	10.1.1.0	-	fa0/1
R	192.168.1.0	10.1.1.1	

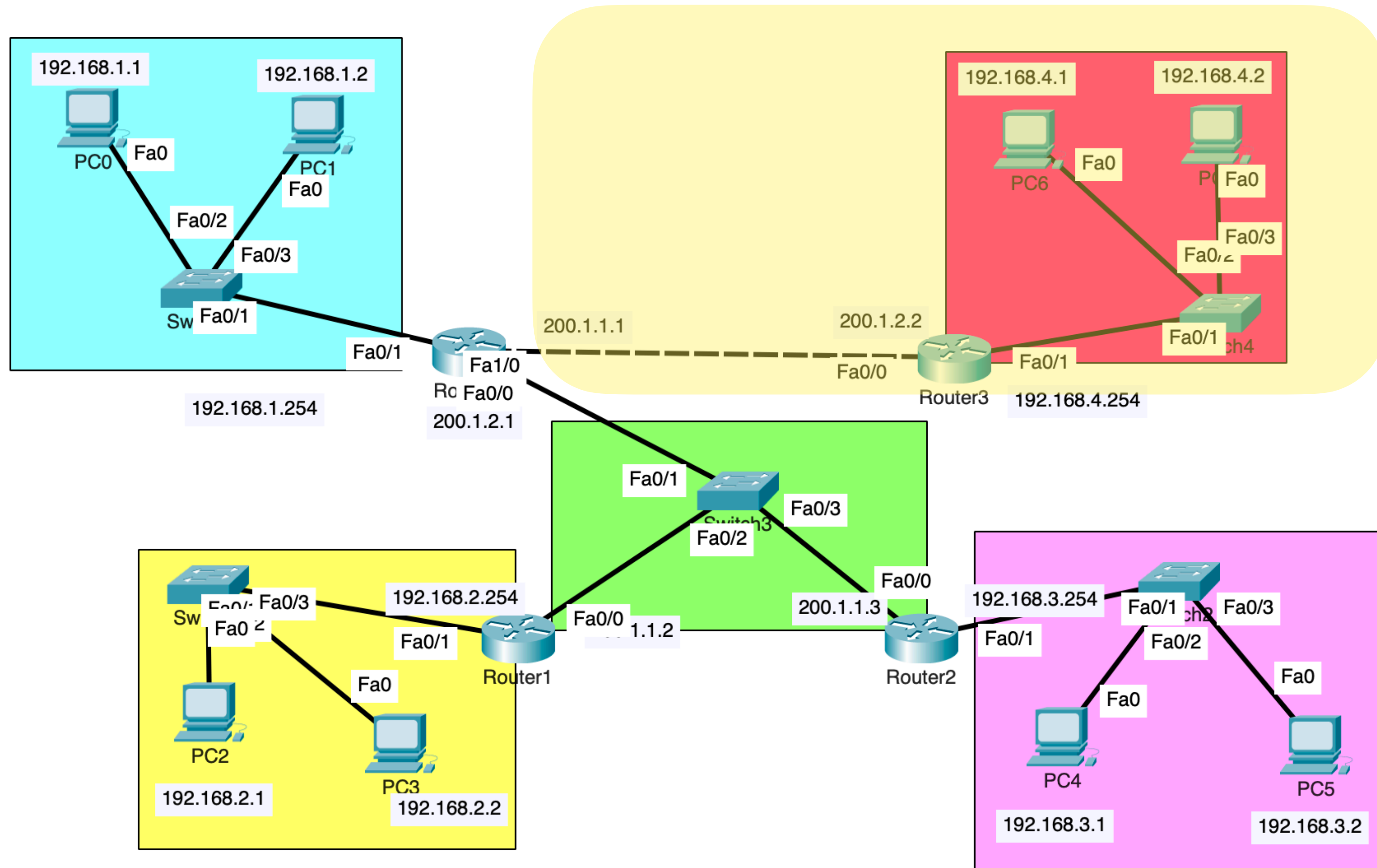
Router1's Table

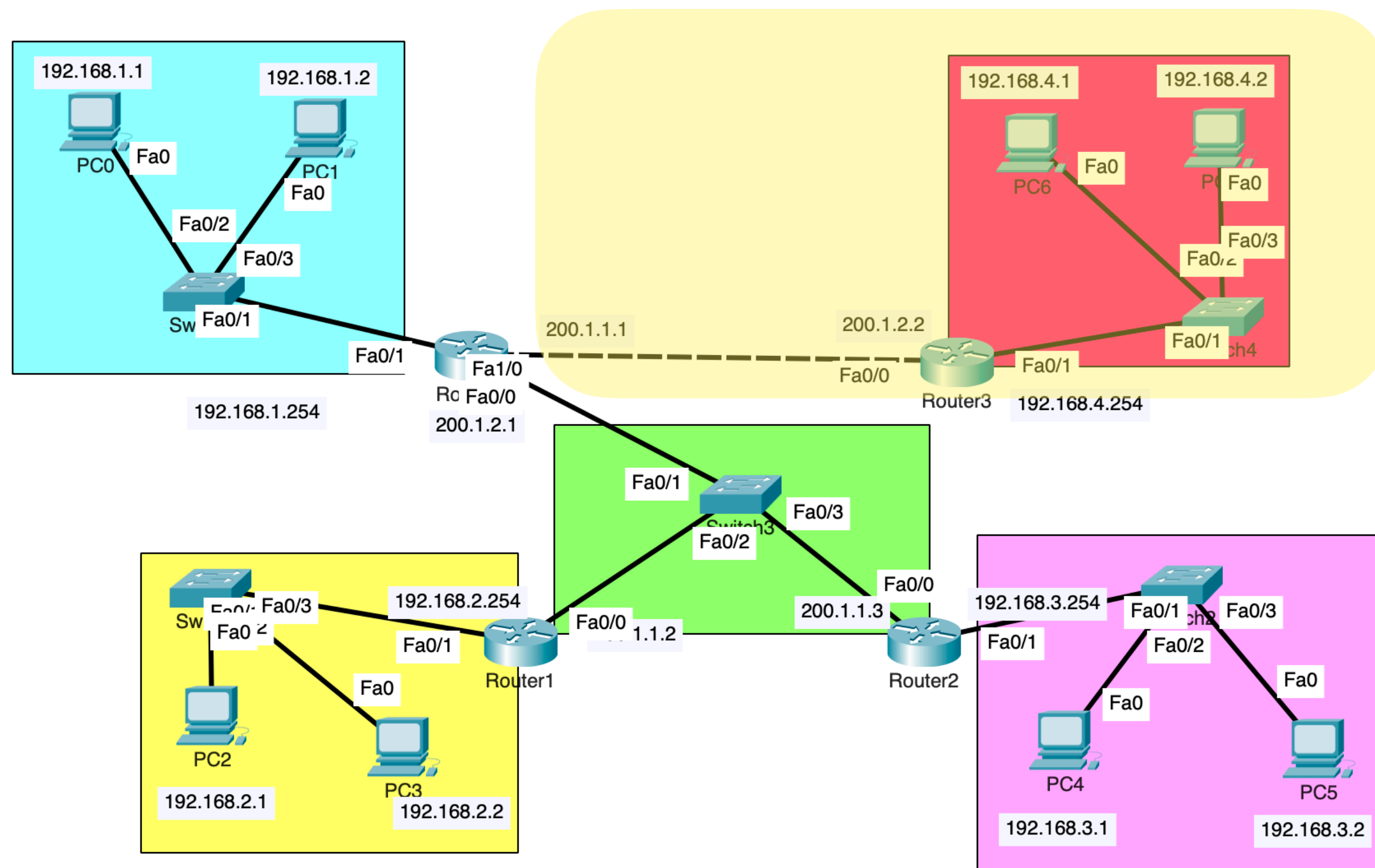
Default Route

Default Route

- Default route is the gateway of last resort. If destination IP can't find a matching destination network in the routing table, the data will go the way of default route
- Default route is a special static route using 0.0.0.0 as the destination network address and 0.0.0.0 as the next hop IP
- Command in Cisco router:

```
Router(config)#ip route 0.0.0.0 0.0.0.0 gateway_IP
```



Add a new network

We need to add a static route on all old routers to this new network

We need to set static route on new router to all old networks

Or we can use default route to simplify the configurations

Router0

```
ip route 192.168.2.0 255.255.255.0 200.1.1.2
ip route 192.168.3.0 255.255.255.0 200.1.1.3
```

Router1

```
ip route 192.168.1.0 255.255.255.0 200.1.1.1
ip route 192.168.3.0 255.255.255.0 200.1.1.3
```

Router2

```
ip route 192.168.1.0 255.255.255.0 200.1.1.1
ip route 192.168.2.0 255.255.255.0 200.1.1.2
```

Router3

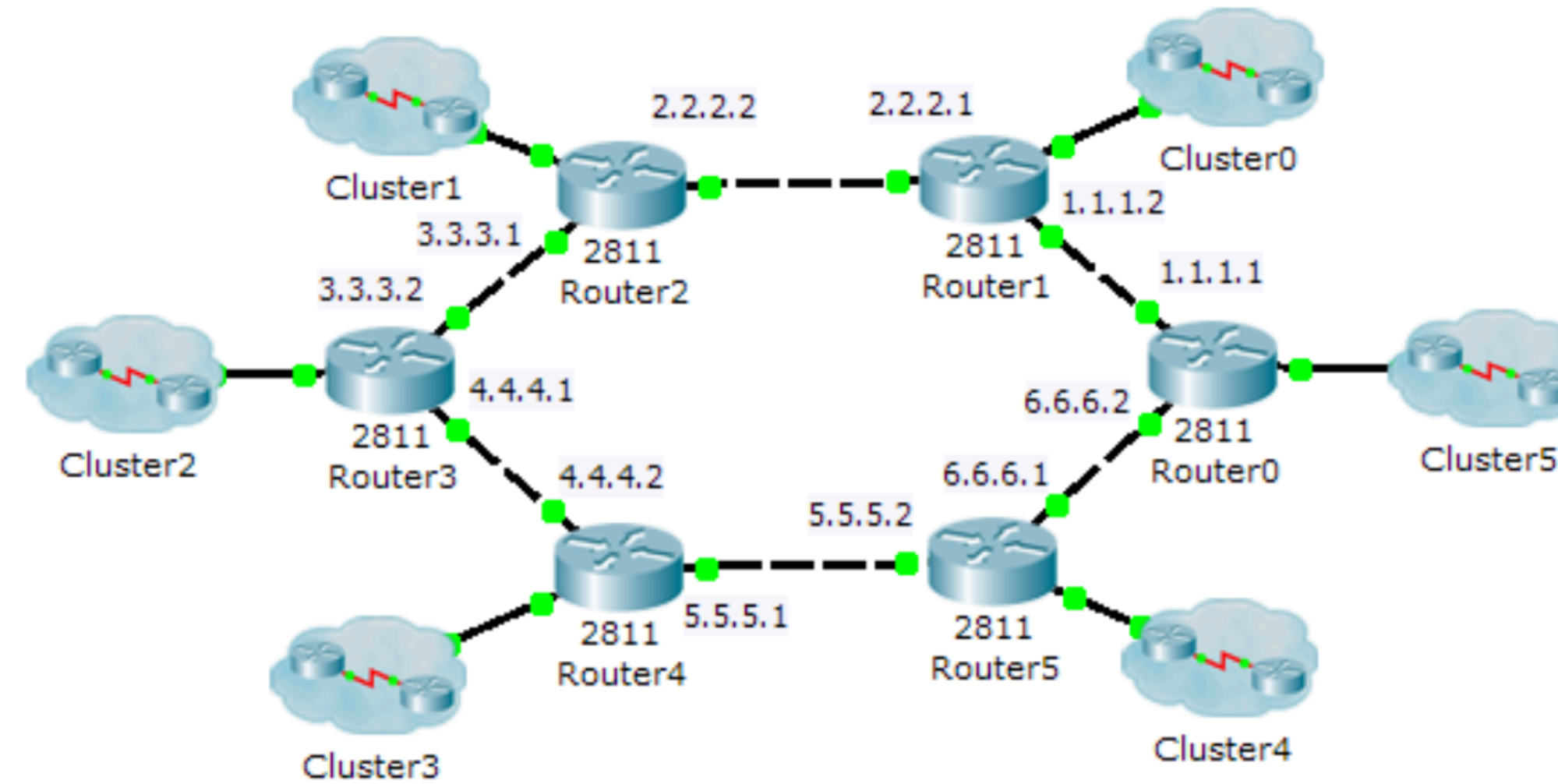
```
ip route 192.168.1.0 255.255.255.0 200.1.1.1
ip route 192.168.2.0 255.255.255.0 200.1.1.2
```

Ring Network

- Ring network can have routing set by setting default routes either in:
 - Counter-clockwise
 - Clock wise

Counter-clockwise routes

```
Router0(config)#ip route 0.0.0.0 0.0.0.0 1.1.1.2  
Router1(config)#ip route 0.0.0.0 0.0.0.0 2.2.2.2  
Router2(config)#ip route 0.0.0.0 0.0.0.0 3.3.3.2  
Router3(config)#ip route 0.0.0.0 0.0.0.0 4.4.4.2  
Router4(config)#ip route 0.0.0.0 0.0.0.0 5.5.5.2  
Router5(config)#ip route 0.0.0.0 0.0.0.0 6.6.6.2
```



Admin Distance

Route Source	AD
Connected interface	0
Static route	1
Enhanced Interior Gateway Routing Protocol (EIGRP) summary route	5
External Border Gateway Protocol (BGP)	20
Internal EIGRP	90
IGRP	100
OSPF	110
Intermediate System-to-Intermediate System (IS-IS)	115
Routing Information Protocol (RIP)	120
Exterior Gateway Protocol (EGP)	140
On Demand Routing (ODR)	160
External EIGRP	170
Internal BGP	200
Unknown*	255

Cisco router choose the best path based on admin distance on the routing table

S	192.168.8.0/24	[1/0]	via 10.10.10.1
R	192.168.8.0/24	[120/1]	via 10.10.20.3
R	192.168.8.0/24	[120/3]	via 10.10.30.3
O	192.168.8.0/24	[110/100]	via 10.10.40.5
S*	192.168.0.0/24	via 10.10.50.5	

Router0’s Table

Router Precedence

- 1.Router always choose static route as first choice
- 2.If there are no static route, then choose the dynamic route with lower admin cost. for example, OSPF over RIPv1
- 3.If there is no route to the destination, send the data to default gateway